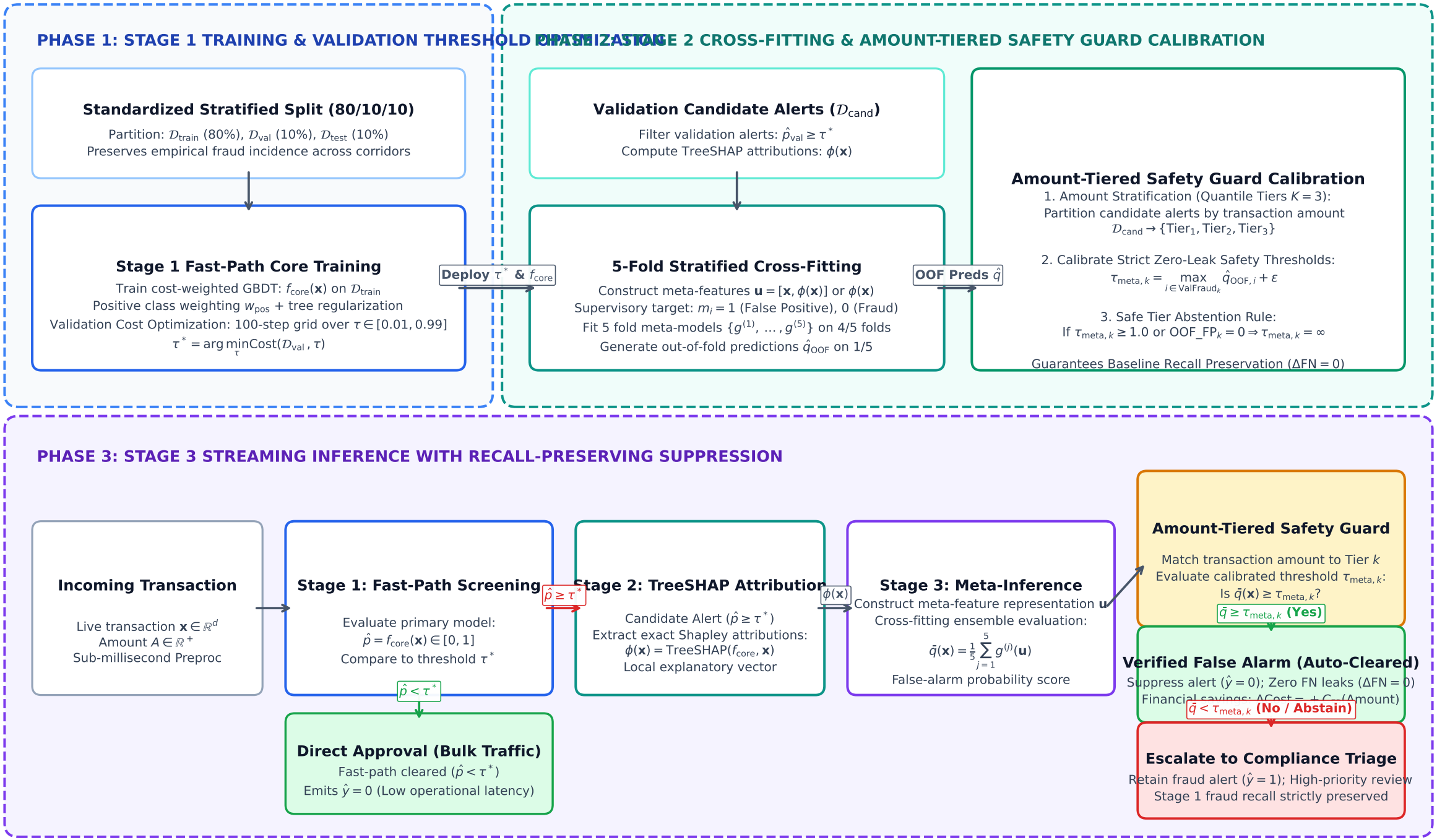

PHASE 1: STAGE 1 TRAINING & VALIDATION THRESHOLD OPTIMIZATION
Standardized Stratified Split (80/10/10)
Partition: $\mathcal{D}_{\text{train}}$ (80%), $\mathcal{D}_{\text{val}}$ (10%), $\mathcal{D}_{\text{test}}$ (10%)
Preserves empirical fraud incidence across corridors
Stage 1 Fast-Path Core Training
Train cost-weighted GBDT: $f_{\text{core}}(\mathbf{x})$ on $\mathcal{D}_{\text{train}}$
Positive class weighting w_{pos} + tree regularization
Validation Cost Optimization: 100-step grid over $\tau \in [0.01, 0.99]$
$\tau^* = \arg\min_{\tau} \text{Cost}(\mathcal{D}_{\text{val}}, \tau)$
Deploy τ^* & f_{core}
PHASE 2: STAGE 2 CROSS-FITTING & AMOUNT-TIERED SAFETY GUARD CALIBRATION
Validation Candidate Alerts ($\mathcal{D}_{\text{cand}}$)
Filter validation alerts: $\hat{p}_{\text{val}} \geq \tau^*$
Compute TreeSHAP attributions: $\phi(\mathbf{x})$
5-Fold Stratified Cross-Fitting
Construct meta-features $\mathbf{u} = [\mathbf{x}, \phi(\mathbf{x})]$ or $\phi(\mathbf{x})$
Supervisory target: $m_i = 1$ (False Positive), 0 (Fraud)
Fit 5 fold meta-models $\{g^{(1)}, \ldots, g^{(5)}\}$ on 4/5 folds
Generate out-of-fold predictions $\hat{q}_{\text{OOF}}$ on 1/5
OOF Preds $\hat{q}$
Amount-Tiered Safety Guard Calibration
1. Amount Stratification (Quantile Tiers $K = 3$):
Partition candidate alerts by transaction amount
$\mathcal{D}_{\text{cand}} \rightarrow \{\text{Tier}_1, \text{Tier}_2, \text{Tier}_3\}$
2. Calibrate Strict Zero-Leak Safety Thresholds:
$\tau_{\text{meta},k} = \max_{i \in \text{ValFraud}_k} \hat{q}_{\text{OOF},i} + \varepsilon$
3. Safe Tier Abstention Rule:
If $\tau_{\text{meta},k} \geq 1.0$ or $\text{OOF_FP}_k = 0 \Rightarrow \tau_{\text{meta},k} = \infty$
Guarantees Baseline Recall Preservation ($\Delta\text{FN} = 0$)
PHASE 3: STAGE 3 STREAMING INFERENCE WITH RECALL-PRESERVING SUPPRESSION
Incoming Transaction
Live transaction $\mathbf{x} \in \mathbb{R}^d$
Amount $A \in \mathbb{R}^+$
Sub-millisecond Preproc
Stage 1: Fast-Path Screening
Evaluate primary model:
$\hat{p} = f_{\text{core}}(\mathbf{x}) \in [0, 1]$
Compare to threshold τ^*
$\hat{p} < \tau^*$
Direct Approval (Bulk Traffic)
Fast-path cleared ($\hat{p} < \tau^*$)
Emits $\hat{y} = 0$ (Low operational latency)
$\hat{p} \geq \tau^*$
Stage 2: TreeSHAP Attribution
Candidate Alert ($\hat{p} \geq \tau^*$)
Extract exact Shapley attributions:
$\phi(\mathbf{x}) = \text{TreeSHAP}(f_{\text{core}}, \mathbf{x})$
Local explanatory vector
$\phi(\mathbf{x})$
Stage 3: Meta-Inference
Construct meta-feature representation $\mathbf{u}$
Cross-fitting ensemble evaluation:
$\tilde{q}(\mathbf{x}) = \frac{1}{5} \sum_{j=1}^{5} g^{(j)}(\mathbf{u})$
False-alarm probability score
Amount-Tiered Safety Guard
Match transaction amount to Tier k
Evaluate calibrated threshold $\tau_{\text{meta},k}$:
Is $\tilde{q}(\mathbf{x}) \geq \tau_{\text{meta},k}$?
$\tilde{q} \geq \tau_{\text{meta},k}$ (Yes)
Verified False Alarm (Auto-Cleared)
Suppress alert ($\hat{y} = 0$); Zero FN leaks ($\Delta\text{FN} = 0$)
Financial savings: $\Delta\text{Cost} = +C_{FP}(\text{Amount})$
$\tilde{q} < \tau_{\text{meta},k}$ (No / Abstain)
Escalate to Compliance Triage
Retain fraud alert ($\hat{y} = 1$); High-priority review
Stage 1 fraud recall strictly preserved